Epileptic Seizure Detection Using Machine Learning: A Comparative Study of Class Imbalance Handling Techniques

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***Abstract*—** Epilepsy affects over 50 million individuals worldwide, with seizures arising from abnormal electrical brain activity. Automated seizure detection from electroencephalogram (EEG) signals is clinically critical but severely impeded by extreme class imbalance—seizure events constitute less than 2% of continuous EEG data, yielding imbalance ratios as high as 115.7:1 in real-world recordings. This paper presents a systematic comparative evaluation of five class imbalance handling strategies on the CHB-MIT Scalp EEG Database under a cross-patient protocol. Eight statistical time-domain features are extracted from non-overlapping 2-second windows across 23 EEG channels, producing 184-dimensional feature vectors compressed by PCA retaining 120 components. AUC-ROC and false detection rate per hour (FDR/hour) are reported alongside F1-score for clinically meaningful evaluation. Statistical robustness is validated across five independent random seeds. The Baseline Random Forest achieves the best AUC-ROC of 0.980 ± 0.001 and a clinically acceptable FDR/hour of 12.91 ± 1.56, significantly outperforming all resampling approaches on both metrics. An ablation study confirms that 120 PCA components is the optimal dimensionality. All models achieve inference times below 0.025ms per window, confirming real-time deployment feasibility.

Keywords— *EEG, epileptic seizure detection, class imbalance, SMOTE, Random Forest, XGBoost, PCA, CHB-MIT, AUC-ROC, false detection rate, real-time detection*

I. Introduction

Epilepsy is one of the most prevalent chronic neurological disorders globally, affecting more than 50 million individuals of all ages [1]. The condition is defined by recurrent, unprovoked seizures arising from abnormal electrical discharges in the brain. Beyond the immediate dangers of injury and sudden unexpected death in epilepsy (SUDEP), the chronic unpredictability of seizures restricts driving, employment, and independent living, profoundly impacting quality of life [2].

Electroencephalography (EEG) is the established clinical gold standard for diagnosing epileptic activity, recording the brain's electrical potentials over extended periods. In practice, long-term EEG monitoring generates hours to days of continuous multi-channel data. Manual expert review is time-consuming, subjective, and subject to significant inter-rater variability—limitations that create a strong clinical need for reliable automated seizure detection systems [3].

A fundamental challenge is the severe class imbalance inherent in continuous EEG. Seizure events account for less than 2% of recording time, producing non-seizure to seizure ratios as high as 115.7:1 in this study. Standard classifiers trained on such data are biased toward the majority class, achieving high overall accuracy but critically insufficient seizure sensitivity. Moreover, common evaluation metrics such as F1-score are threshold-dependent, while AUC-ROC provides a threshold-independent assessment of discriminative ability. Equally important for clinical deployment is the false detection rate per hour (FDR/hour), which directly quantifies alert burden on clinical staff.

This paper makes the following contributions: (i) the first unified comparison of five class imbalance strategies for EEG seizure detection under cross-patient evaluation using both AUC-ROC and FDR/hour as clinical evaluation metrics; (ii) statistical robustness validation across five random seeds; (iii) an ablation study on PCA component count; (iv) decision threshold tuning analysis reporting FDR/hour at each operating point; and (v) real-time feasibility analysis confirming sub-millisecond inference for all models.

II. Related Work

The CHB-MIT Scalp EEG Database, introduced by Shoeb and Guttag [4], is the de facto benchmark for automated seizure detection. Shoeb and Guttag proposed a patient-specific SVM-based detector, demonstrating high within-patient sensitivity and establishing the evaluation conventions adopted in most subsequent studies. Subasi and Gursoy [5] showed that PCA-compressed statistical features with SVM yield competitive inter-patient performance, motivating the feature-PCA pipeline in the present work.

Statistical time-domain features including mean, variance, skewness, kurtosis, and RMS have consistently proven discriminative for seizure classification [6]. Acharya et al. [7] reviewed EEG feature extraction and found that combinations of time-domain and frequency-domain features consistently outperform either group alone, supporting the eight-feature set used here. Chawla et al. [8] introduced SMOTE for minority class oversampling; however, at extreme imbalance ratios its default 1:1 strategy inflates the feature space with synthetic samples, collapsing classifier precision as observed in this study.

Batista et al. [9] proposed SMOTETomek as a hybrid approach combining SMOTE with Tomek Links boundary

cleaning. Chen et al. [10] introduced the Balanced Random Forest, undersampling the majority class within each bootstrap draw. Chen and Guestrin [12] introduced XGBoost with scale_pos_weight that encodes imbalance into the gradient boosting loss. Despite availability of these techniques, a unified comparison incorporating AUC-ROC, FDR/hour, and statistical robustness validation on EEG data remains absent from the literature.

III. Dataset and Methodology

Figure 1: Proposed Methodology Pipeline



Fig. 1. Proposed methodology pipeline for EEG-based seizure detection.

A. Dataset

The CHB-MIT Scalp EEG Database [4] comprises long-term EEG recordings from 22 paediatric subjects at Children's Hospital Boston, with 23 channels sampled at 256 Hz and expert-annotated seizure times. Patients chb01 and chb03 are used for training (21,588 windows; 185 seizure, 21,403 non-seizure); patient chb05 files chb05_06.edf and chb05_13.edf form the held-out test set (3,598 windows; 113 seizure, 3,485 non-seizure). The training class imbalance ratio is 115.7:1.

B. Feature Extraction

EEG recordings are segmented into non-overlapping 2-second windows and labelled as ictal or interictal. For each window and each of the 23 channels, eight statistical features are extracted as summarised in TABLE V, yielding a 184-dimensional (23×8) feature vector. Features are normalised using StandardScaler prior to dimensionality reduction.

TABLE V
EXTRACTED STATISTICAL FEATURES PER EEG CHANNEL

Feature	Description
Mean	Average amplitude of EEG window
Variance	Spread of signal amplitude values
RMS	Root mean square; reflects signal power
Skewness	Asymmetry of amplitude distribution
Kurtosis	Tailedness; elevated during seizures
Maximum	Peak positive amplitude in window
Minimum	Peak negative amplitude in window
Band Power	Total spectral power across EEG bands

C. Dimensionality Reduction

PCA is applied to the normalised features. The PCA transform is fitted exclusively on training data and applied as a fixed projection to test features to prevent information leakage. The optimal component count of 120 is validated through the ablation study reported in Section IV-D.

D. Class Imbalance Handling Techniques

Five strategies are evaluated: (1) Baseline Random Forest (300 estimators, max depth 12); (2) SMOTE with sampling_strategy=0.3; (3) Balanced Random Forest with majority undersampling per bootstrap; (4) SMOTETomek combining SMOTE with Tomek Links boundary cleaning; and (5) XGBoost with scale_pos_weight=115.7 encoding the imbalance ratio into the loss function.

E. Evaluation Metrics

Four metrics are reported for the seizure class: Precision, Recall, F1-score, and AUC-ROC. Additionally, the False Detection Rate per hour (FDR/hour) is computed as the number of false positive seizure predictions divided by total test recording hours. FDR/hour is the metric most relevant to clinical deployment, directly quantifying alert burden. All models are evaluated across five random seeds (42, 123, 456, 789, 999) for statistical robustness.

IV. Results and Discussion

A. Technique Comparison

TABLE I reports seizure-class metrics for all five techniques on the held-out patient chb05. The Baseline RF achieves the best AUC-ROC (0.979) and lowest FDR/hour (14.01), indicating that despite moderate F1 at the default threshold, the model has excellent discriminative ability and generates only 14 false alarms per hour of monitoring. SMOTE and SMOTETomek achieve identical results (AUC-ROC=0.894, FDR/hour=98.55), indicating SMOTETomek provides no additional benefit over SMOTE at this imbalance level. The Balanced RF achieves the highest recall (0.991)—detecting virtually every seizure—but at a clinically catastrophic FDR/hour of 515.29, equivalent to more than 8 false alarms per minute, which would cause severe alert fatigue.

TABLE I
COMPLETE SEIZURE CLASS EVALUATION (SEED=42). BOLD = BEST PER COLUMN.

Technique	Prec.	Recall	F1	AUC-ROC	FDR/hr
Baseline RF	0.705	0.593	**0.644**	**0.979**	**14.01**
SMOTE+RF	0.117	0.230	0.155	0.894	98.55
Balanced RF	0.098	0.991	0.178	0.904	515.29
SMOTETomek+R	0.117	0.230	0.155	0.894	98.55

F					
XGBoost(SPW)	0.178	0.708	0.285	0.916	184.60

B. Statistical Robustness

TABLE II reports Baseline RF performance across five independent random seeds. The AUC-ROC is remarkably stable at 0.980 ± 0.001, confirming that the model's discriminative ability is consistent and not seed-dependent. F1-Score varies more (0.677 ± 0.031) as it is sensitive to the classification threshold. FDR/hour of 12.91 ± 1.56 is clinically acceptable across all seeds.

TABLE II
BASELINE RF STATISTICAL ROBUSTNESS (5 RANDOM SEEDS)

Metric	Mean	Std Dev
Precision	0.733	0.030
Recall	0.628	0.036
F1-Score	0.677	0.031
AUC-ROC	0.980	0.001
FDR/hour	12.910	1.562

C. Threshold Tuning

TABLE III presents threshold tuning on the Baseline RF. At threshold 0.40, the best F1-score of 0.662 is achieved with Recall=0.894; however, FDR/hour rises to 45.53. At threshold 0.50 (default), the model achieves the best FDR/hour (14.01) with F1=0.651. This illustrates the fundamental precision-recall-FDR tradeoff: clinicians can adjust the threshold to either minimise missed seizures (lower threshold) or minimise false alarms (higher threshold) depending on the clinical context.

TABLE III
THRESHOLD TUNING ON BASELINE RANDOM FOREST

Threshold	Precision	Recall	F1-Score	FDR/hour
0.50	0.708	0.602	**0.651**	14.01
0.40	0.526	0.894	**0.662**	45.53
0.30	0.372	0.938	0.533	89.55
0.20	0.276	0.973	0.431	144.08
0.15	0.234	0.973	0.377	180.60
0.10	0.185	0.973	0.312	241.63

D. Ablation Study: PCA Components

TABLE IV validates the choice of 120 PCA components. Below 120, both F1 and AUC-ROC decrease monotonically (F1=0.474 at 60 components), confirming that additional components contribute discriminative information. At 150 components (100% variance explained), F1 drops to 0.567 while AUC-ROC remains 0.979—indicating overfitting to noise in the last few components. The 120-component configuration at 99.6% variance explained represents the optimal dimensionality-accuracy tradeoff.

TABLE IV
PCA COMPONENT ABLATION STUDY (BASELINE RF, SEED=42).
BOLD = SELECTED.

PCA Comp.	Var. Expl. (%)	F1-Score	AUC-ROC
40	87.8	0.328	0.951
60	94.4	0.474	0.964
90	98.6	0.581	0.976
120 (selected)	99.6	**0.644**	**0.979**
150	100.0	0.567	0.979

E. Real-Time Feasibility

All five models achieve inference times below 0.025ms per 2-second window, far below the 2000ms real-time threshold. XGBoost is the fastest at 0.006ms/window. These results confirm that all evaluated approaches are suitable for real-time deployment in continuous EEG monitoring systems without requiring specialised hardware.

F. Discussion

The key finding of this study is that AUC-ROC is a more reliable metric than F1-score for evaluating seizure detectors under extreme class imbalance. While F1-scores across models range from 0.155 to 0.644—suggesting large performance differences—AUC-ROC values range from 0.894 to 0.979, showing that all models have reasonable discriminative ability. The Baseline RF achieves an AUC-ROC of 0.979, indicating that given appropriate threshold selection, it can achieve excellent clinical performance.

The FDR/hour analysis reveals why resampling techniques fail clinically despite improving recall: SMOTE achieves Recall=0.230 with FDR/hour=98.55; Balanced RF achieves Recall=0.991 but FDR/hour=515.29. In contrast, Baseline RF at the default threshold achieves FDR/hour=14.01—the only clinically acceptable rate among all evaluated configurations. These results demonstrate that aggressively maximising recall through resampling creates clinically unusable detectors.

V. Conclusion

This paper presented a systematic comparison of five class imbalance handling techniques for EEG-based epileptic seizure detection on the CHB-MIT dataset, evaluated under a cross-patient protocol with AUC-ROC, FDR/hour, and F1-score reported across five random seeds. The Baseline Random Forest achieves the best overall performance: AUC-ROC of 0.980 ± 0.001 and FDR/hour of 12.91 ± 1.56—the only configuration producing a clinically acceptable false alarm rate. All models achieve inference times below 0.025ms per window, confirming real-time feasibility.

Three key conclusions emerge: (i) AUC-ROC is a more reliable metric than F1-score for imbalanced EEG evaluation, as it is threshold-independent and captures discriminative ability across all operating points; (ii) resampling techniques dramatically increase FDR/hour to clinically unacceptable levels (98–515 false alarms/hour), making them unsuitable for clinical deployment despite recall improvements; and (iii) the ablation study validates 120 PCA components as the optimal dimensionality for this task.

Future work will extend the patient cohort to all 22 CHB-MIT subjects with leave-one-patient-out cross-validation, investigate patient-adaptive threshold calibration, and explore deep learning architectures with focal loss for improved minority class representation.

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